

Personalized AI assistants could either promote or hinder the emergence of collective intelligence

Jaeseung Kim¹ Sang Woo Park^{1,2,3,*}

¹ School of Biological Sciences, Seoul National University, Seoul, Korea

² Department of Ecology and Evolution, University of Chicago, Chicago, IL, USA

³ Department of Ecology and Evolutionary Biology, Princeton University, Princeton, NJ, USA

*Corresponding author: sangwoopark@snu.ac.kr

Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI services on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI services on human CI is hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of indentifying a robust incentive structure that can promote CI in the presence of AI, which can retain the benefits of AIs while preventing potential risk such as preserving the diversity of individual opinions.

To address these questions, we extended previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. We allowed each individual to independently consult with personal AIs and revise their predictions accordingly. Across different assumptions about the AI model’s ability and underlying incentive structures, we investigate the robustness of human CIs.

Main

Model

We begin by extending previous models of CI in which a group of players makes collective predictions by gathering information and aggregating individual predictions in a decentralized manner [30, 12, 13]. In the previous approach, among M different factors (x_i) that affect the outcome (Y), each individual makes a prediction based on their factor of interest (x_{e_A}) and their own belief (c_A): $Y_A = c_A x_{e_A}$. Here, we assume each individual can independently consult AI and revise their prediction based on their reliance on AI (β_A): $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$.

To specify how individuals consult AI, we model the AI prediction (Y_A^{AI}) as a noisy and biased estimate of the target outcome. Specifically, AI predictions include factor-specific bias (b_i) and stochastic error (ϵ). We then consider two extreme modes of AI response, which differ in the amount of information AI uses when consulting an individual user. The first is the “AI knows all” model, in which AI predicts the outcome based on all factors regardless of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$, where α_i represents the true coefficient of the factor. The second is the “AI answers question” model, in which AI predicts the outcome based only on the user’s interest: $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$.

After the prediction, each individual is rewarded by the expected value of their payoff under the incentive structure. We adopt two incentive structures, “feedback” and “niche expert” from Wang et al. [13], which have been shown to promote near-perfect collective accuracy without AI (except averaging aggregation with niche expert incentive). The feedback incentive gives higher payoff to individuals who can push the collective prediction toward the true outcome, regardless of their individual predictive accuracy. By contrast, the niche expert incentive gives higher payoff to individuals who have both better predictive accuracy and a minority interest.

Through time, the group evolves by imitating the peers who got better payoff. Specifically, for each generation, two individuals are randomly sampled, and one imitates all three strategies (factor of interest, belief, reliance on AI) of the other with a probability determined by the difference in expected payoff. Individual predictions are aggregated into a collective prediction by simple averaging in the AI knows all model, or by clustering in the AI answers question model, depending on the mode of AI.

AI can both promote or prevent the emergence of CI

To understand how AI could affect the emergence of CI, we compare collective dynamics with and without AI assistance. We find that AI affects CI in qualitatively different ways depending on the mode of AI.

In the AI knows all model, AI promotes CI by increasing either the speed at which CI emerges or the resulting collective accuracy (Fig. 2A). When the popula-

tion can achieve near-perfect CI without AI (feedback incentive), AI accelerates its emergence by raising the initial collective accuracy to the level of AI accuracy. By contrast, when the population cannot achieve CI without AI (niche expert incentive), AI enables moderately accurate CI that exceeds the accuracy achievable by either AI or the human population alone. These trends are robust across different levels of AI accuracy and can be shown mathematically by assuming a sufficiently large population of players (see SI).

In contrast, In the AI answers question model, AI prevents the emergence and maintenance of CI (Fig. 2B). Under the feedback incentive, the population initially achieves near-perfect CI. However, as interest diversity rapidly decreases, the population fails to capture variation in some factors, leading to increased collective variance and a breakdown of collective accuracy. Under the niche expert incentive, collective accuracy initially increases but peaks before near-perfect CI is achieved, and then plateaus with a slight decline. This trend is mediated by increased reliance on AI: individuals with higher reliance on AI obtain higher payoffs, causing the population to become increasingly reliant on AI and the collective prediction to inherit AI bias. However, the niche expert incentive can also drive reliance on AI toward zero, depending on the relationship between the true coefficient and AI bias. In this case, the population excludes AI assistance and builds CI on its own, which largely delays the emergence of CI (see SI).

Directly incentivizing or penalizing AI usage

Because AI use can strongly influence group performance, one possible intervention is to directly incentivize or penalize AI use itself. To investigate whether such direct incentives or penalties improve CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$). Except in the AI knows all model under the feedback incentive, where near-perfect CI can be achieved regardless of the incentive structure, directly incentivizing or penalizing AI use generally leads to lower collective accuracy than the original incentive structures (Fig. 3).

When AI use is directly incentivized, reliance on AI evolves toward full reliance ($\beta_A = 1$), reducing collective accuracy as the collective prediction inherits AI bias. By contrast, when AI use is directly penalized, collective accuracy decreases with the strength of the penalty. In some cases, when AI bias is relatively large and the penalty is weak, penalizing AI use can achieve higher collective accuracy than the original incentive structure. However, in most cases, penalizing AI use imposes strong selection against AI reliance, pushing the population toward lower stationary collective accuracy. Although the AI answers question model under the feedback incentive already shows stochastic breakdown of CI due to the loss of interest diversity,

142 incentivizing or penalizing AI use further strengthens the breakdown.

143 **Balanced incentives mitigate AI-induced failures of CI**

144 Focusing on the AI answers question model, in which both incentive structures fail
145 to promote accurate CI, we investigate whether a better incentive structure can
146 overcome the risks introduced by AI. We propose a “balanced” incentive, defined as
147 the simple average of the feedback and niche expert incentives.

148 The core idea of the balanced incentive is to maintain both within and between
149 interest diversity. The feedback incentive can strongly maintain within interest diver-
150 sity, defined as diversity in belief among individuals with the same factor of interest.
151 However, between interest diversity can be hindered when AI assistance is present.
152 This occurs because revising individual predictions with AI predictions compresses
153 the belief distribution of the population. Although the initial belief distribution is
154 shifted toward the true value, the compressed distribution makes it harder for the
155 population to achieve near-perfect predictions through social learning for important
156 factors (factors with large absolute coefficients). This eventually leads to a strong
157 flow of players from relatively less important factors to more important factors, caus-
158 ing a loss of between interest diversity.

159 By contrast, the niche expert incentive can maintain between interest diversity be-
160 cause it directly incentivizes minority interests. However, it cannot maintain within
161 interest diversity because, within the same interest, individuals are rewarded only ac-
162 cording to their individual predictive performance. This leads to strong selection for
163 either high or low reliance on AI, which in turn results in lower collective accuracy.

164 The balanced incentive can promote near-perfect CI by maintaining both within
165 and between interest diversity, regardless of AI accuracy. Reliance on AI decreases as
166 AI accuracy decreases, reflecting the effective use of AI assistance. However, human
167 accuracy, defined as collective accuracy without AI, remains largely stochastic, and
168 only moderate accuracy can be achieved without AI.

169 **Discussion**

170 We present an epidemiological analysis of RSV outbreaks in Japan combining spa-
171 tiotemporal observations with dynamical disease modeling. Our analysis revealed
172 semiannual cycles in seasonal RSV transmission in four major islands (Honshu,
173 Shikoku, Kyushu, and Ryukyu), which correlate with specific humidity and tem-
174 perature. We found that these semiannual cycles allowed a sudden shift in the
175 seasonality of RSV outbreaks in response to an increase in RSV transmission—this
176 increase could be captured by either through changes in susceptibility or transmissi-
177 bility. We hypothesize that an increase in childcare capacity through Comprehensive
178 Support System for Children and Childcare may have contributed to the increase in
179 RSV transmission [?].

180 Our analysis revealed considerable heterogeneity in epidemic dynamics across
181 major islands in Japan. We showed that these differences could be explained by
182 the differences in underlying seasonal transmission. Notably, we found a robust,
183 quadratic relationship between the estimated transmission rates against mean spe-
184 cific humidity and mean temperature across four major islands (Honshu, Shikoku,
185 Kyushu, and Ryukyu), indicating low transmission at intermediate levels of specific
186 humidity. These findings echo earlier studies that demonstrated similar relationships
187 for RSV [?] and influenza [? ? ? ? ?]. However, given strong correlation between
188 specific humidity and temperature, we were unable to identify the primary environ-
189 mental driver from the current study. Their relative contributions may be separated
190 from data spanning a wider geographical region with more climate variability as well
191 as epidemiological variability [?].

192 The robustness of this quadratic relationship across islands exhibiting different
193 climate conditions also suggests a possibility that climate-driven transmission may
194 be, in part, facilitated by human behavior: for example, an increase in time spent
195 indoors during low and high humidity seasons can contribute to increased transmis-
196 sion. Further investigation is needed to establish the underlying mechanism behind
197 how climate conditions affect the transmission of respiratory pathogens, especially
198 across different environmental contexts (e.g., indoor vs outdoor). We note that this
199 relationship is correlational, rather than causal, and therefore any other climate vari-
200 ables (e.g., rainfall) or seasonal variation in human behavior (e.g., school terms) that
201 correlate with seasonal variation in specific humidity and temperature will be implic-
202 itly captured by this relationship. Understanding why the relationship between RSV
203 transmission and humidity in Hokkaido island differs from other locations remains
204 to be answered.

205 We hypothesized that the Comprehensive Support System for Children and Child-
206 care may have contributed to the increase in RSV transmission, but other mecha-
207 nisms may have also contributed. For example, an increase in inbound overseas trav-
208 elers may have led to an increased exposure to RSV among adults thereby increasing
209 the total amount of transmission [?]. However, RSV typically infects young chil-
210 dren [?] and contact rates among young children and adolescents (5–19) are much
211 higher than among older adults in Japan [?], suggesting that an increase in
212 inbound overseas travelers may have smaller effects than an increase in childcare at-
213 tendance on overall RSV dynamics. Another competing hypothesis that could lead
214 to an increase in transmission would be strain evolution: for example, one study
215 noted that L172Q/S173L mutant strains of RSV B that became dominant around
216 2016 had reduced susceptibility against monoclonal antibodies [?]. However, it is
217 not yet clear how this mutation translates to susceptibility against infection-derived
218 antibodies. While our findings are consistent with those by [?], who also pointed
219 out the association between an increase in childcare attendance and a large change
220 in the seasonality of Kawasaki disease in Japan, we cannot rule out the possibility
221 that other factors, such as increase in air travel [?] or strain evolution [?], could
222 have contributed to the increase in overall transmission.

Our sensitivity analysis revealed that an increase in transmissibility, rather than an increase in susceptibility, can also explain a sudden shift in seasonality in RSV outbreaks. In fact, both models provide evidence for increased contact rates. Specifically, the total rate of RSV transmission can be written as $\hat{\beta}cSI/N$, where $\hat{\beta}$ represents the rate of transmission per contact, c represents the contact rate, S represents the number of susceptible individuals, I represents the number of infected individuals, and N represents the total population size. Then, an increase in contact rate c can be captured by either increase in S (the original model) and $\hat{\beta}$ (the sensitivity analysis). Therefore, conclusions from both models are consistent with our hypothesis that an increase in contact rates, especially among children, contributed to the sudden change in RSV seasonality. Detailed age structured surveillance data, alongside age structured model, will be needed to properly assess the relative contribution of potential mechanisms, such as increase in childhood mixing patterns or increase in inbound overseas travelers.

Our model-based estimate of the initial susceptible fraction range between 8%–11%. However this estimate must be interpreted with care as it represents effective changes in population-level susceptibility, which can depend on many factors such as immunological and behavioral changes, rather than an actual susceptible fraction. Therefore, these estimates are not directly comparable to serological estimates of proportion seronegative. For example, [?] recently estimated that $\sim 90\%$ of individuals above 3 years of age have been infected at least once by analyzing anti-pre-F protein antibody concentration data from the Netherlands [? ?], but individuals with RSV antibodies can still permit reinfection. A cross-section serological assay from healthcare and non-healthcare workers at a pediatric medical facility in Japan found that $\sim 95\%$ of the participants were seropositive but only 63.5% had greater than 8 fold neutralizing antibody titers [?], suggesting challenges in translating serological data to susceptible fraction. While our estimates are not inconsistent with these data, detailed age structured serological data and model are needed to accurately estimate changes in population-level susceptibility against RSV.

Interventions to slow the transmission of COVID-19 have disrupted the circulation of many pathogens [? ? ? ?], including RSV epidemics in Japan. This disruption has added major challenges to predicting future outbreaks, which prevented us from making long-term predictions. Continued analysis of RSV dynamics in the post-COVID period, particularly with regard to whether RSV outbreaks in Japan return to fall or winter outbreaks, may help further validate our models.

Our analysis relied on several simplifying assumptions. For example, our model assumed that the waning of immunity can render previously infected individual to become fully susceptible to reinfections; this approximation allowed us to reconstruct the dynamics of susceptible hosts more easily. In practice, immunity is likely more complex with secondary infections being less susceptible and transmissible than primary infections [?]. Other studies have also suggested the importance of interaction between RSV A and B [? ?] as well as competition between RSV and human metapneumovirus [?]; our model did not account for such strain dynamics. We

also did not account for explicit spatial structure or underlying stochasticity of the system. Therefore, our estimates of transmission rates must be interpreted with caution as they may implicitly capture factors that we did not account for explicitly, including strain dynamics, spatial structure, and exogenous transmission. Our analyses also primarily focused on three major islands (Honshu, Shikoku, and Kyushu), necessitating a better understanding of RSV dynamics in Hokkaido and Ryukyu islands; however, we note that these three islands make up $> 95\%$ of the population in Japan, meaning that our analyses capture the majority of RSV transmission in Japan. Despite these limitations, our model likely represents a parsimonious approximation of the complex host-pathogen interactions, allowing us to draw general conclusions about how interactions between endogenous (contact rates) and exogenous (climate-driven factors) factors can give rise to a sudden dynamical transition.

Understanding how endogenous and exogenous factors shape epidemic dynamics is critical to predicting future outbreaks and making public health decisions. Our analysis shows that the interplay between climate-driven transmission and subtle changes in contact rates can cause a sudden transition in epidemic dynamics. More broadly, our analysis demonstrates that detailed epidemiological time series data can allow us to tease apart endogenous and exogenous factors in explaining dynamical transitions, offering unique insights into a long-standing ecological question.

Methods

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target Y is modeled as a linear combination of M independent factors $(x_1, x_2, \dots, x_m)$ and $M + 1$ coefficients $(\alpha_0, \alpha_1, \dots, \alpha_m)$, where α_0 denotes the global offset (Fig. 1A). The M factors are randomly drawn from normal distributions with mean X_i and standard deviation σ_i in each generation.

Among a population of N individuals, each individual can observe a single factor and make a prediction based on the observed value and their own belief. In particular, individual A 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (2)$$

where c_A represents individual A 's personal belief, e_A represents the factor that A observes (interest), and x_{e_A} represents the realized value of A 's interest. Individuals can also choose not to observe any factor, which represents stubborn individuals who make predictions only based on their own belief. For notational convenience, we assume that stubborn individuals have interest 0, and let x_0 always be 1.

Next, we model AI as a prediction system that has learned the true coefficients with bias and error (Fig. 1B). For each factor $i \in [0, M]$, AI knows the exact coefficient α_i and has a fixed bias b_i , resulting in a prediction of $\alpha_i x_i + b_i$. In terms of human-AI interaction, we model two contrasting modes of AI: the "AI knows all"

304 model and the “AI answers question” model. In the AI knows all model, AI acts
 305 as an omniscient system and gives an answer regardless of the individual’s interest,
 306 accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (3)$$

307 In contrast, the AI answers question model acts as a chatbot and gives an answer
 308 only based on the individual’s interest. Thus, given the individual’s interest, e_A , AI
 309 predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (4)$$

310 In both models, $\epsilon \sim N(0, \sigma_e^2)$ represents random error arising from the stochastic
 311 generative process of the AI model.

312 AI works as a personalized assistant for each individual, such that each individual
 313 receives assistance by incorporating the AI’s prediction. However, even with the same
 314 opportunity to receive AI assistance, reliance on AI can vary substantially across
 315 individuals. To model this, we assume that the revised prediction is a weighted sum
 316 of the individual’s own prediction and the AI’s prediction by introducing AI belief
 317 β_A as a weighting coefficient. The revised prediction of individual A can be written
 318 as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (5)$$

319 The revised predictions of individuals are aggregated by two different mechanisms,
 320 depending on the mode of AI. In the AI knows all model, even if an individual can
 321 observe only a single factor, their revised prediction contains information about all
 322 factors through interaction with AI. Therefore, we use simple averaging of revised
 323 predictions as the collective prediction. In contrast, in the AI answers question
 324 model, each individual gives a prediction based only on a single factor even after
 325 revising their prediction with AI assistance. Therefore, we use clustering aggregation
 326 to obtain the collective prediction, which is performed by clustering individuals based
 327 on interest and summing the average values of each cluster.

328 For each generation, each individual is rewarded with a payoff based on the given
 329 incentive structure (Fig. 1C). We adopt two incentive structures, “feedback” and
 330 “niche expert,” which were proposed by Wang et al. [13] and promote collective
 331 intelligence when prediction is performed without AI. The feedback incentive gives
 332 higher payoff to individuals who can push the collective prediction toward the true
 333 value, regardless of their individual performance. Given individual A ’s revised pre-
 334 diction $Y_A^{revised}$ and the collective prediction $\hat{Y}^{revised}$, it can be expressed as:

$$\pi_A = Y_A^{revised} (Y - \hat{Y}^{revised}) \quad (6)$$

335 The niche expert incentive gives higher payoff to individuals who have both better
 336 individual performance and a minority interest. Specifically, the niche expert payoff

337 is expressed as the product of the proportion of individuals who observe the same
 338 factor as individual A , ρ_{e_A} , and individual accuracy. Given individual A 's revised
 339 prediction $Y_A^{revised}$, it can be expressed as:

$$\pi_A = -\rho_{e_A}(Y_A^{revised} - Y)^2 \quad (7)$$

340 These two incentive structures were shown to achieve nearly perfect collective in-
 341 telligence through social learning (except for the niche expert incentive with simple
 342 averaging aggregation) by promoting individual expertise while maintaining collec-
 343 tive diversity. Therefore, we use these incentive structures as a baseline to under-
 344 stand how AI affects the emergence of collective intelligence. Moreover, we modify
 345 the two baseline incentives in a later section to understand the consequences of direct
 346 incentives and penalties for AI usage.

347 Throughout the simulation, individuals engage in social learning by imitating
 348 other individuals who receive higher payoffs. We assume that prediction events
 349 occur more frequently than social learning events within the population, such that
 350 individual payoff can be computed as an expected payoff. The imitation process
 351 is performed for at most one pair in each generation. By sampling two random
 352 individuals, A and B , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (8)$$

353 where s denotes the strength of social learning. When imitation occurs, A imitates
 354 all three strategies of B : interest, belief, and AI belief.

355 To evaluate and analyze the dynamics of collective prediction, we define six sum-
 356 mary metrics: collective accuracy, collective bias, collective variance, human accu-
 357 racy, interest diversity, and median AI belief. These metrics capture the accuracy
 358 of the revised collective prediction, its bias-variance decomposition, the predictive
 359 performance without AI, the diversity of individual interests, and median AI belief
 360 in the population (Table 1).

361 Simulations

362 We run a series of simulations to understand how the interplay between climate-
 363 driven transmission and population-level susceptibility can drive a sudden shift in
 364 the timing of an epidemic. First, we simulated the model for a year (from week 26 of
 365 the starting year to week 25 of the following year) by varying the initial conditions
 366 and computing the center of gravity. Specifically, we varied $i(0)$ between 1×10^{-4}
 367 and 3×10^{-3} and $s(0)$ between 0.078 and 0.105. All other parameters were set to
 368 posterior median estimates.

369 To further understand how the shape of seasonal transmission term affects the
 370 degree of shift in seasonality, we varied the shape of seasonal transmission term by
 371 interpolating the estimated β_{seas} from Honshu and Kyshu islands, which are the two
 372 most populated islands. To do so, we first took posterior median estimates of β_{seas}

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$	Squared systematic deviation of the revised collective prediction from the true value.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{revised}})$	Variability of the revised collective prediction error across prediction events.
Human accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the human-only collective prediction relative to the baseline prediction α_0 .
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median AI belief	$\text{median}_{A \in N}(\beta_A)$	Median value of AI belief β_A in the population.

from two islands and fitted generalized additive model with cyclic cubic spline bases to obtain smoothed estimates of β_{seas} for each island, which we denote as β_H and β_K , respectively. Then, we normalized seasonal transmission rates such that it has a mean of zero and has an amplitude of 1:

$$\zeta_X(t) = \frac{1}{\alpha_X} \left(\frac{\beta_X(t)}{\bar{\beta}_X} - 1 \right) \quad (9)$$

where $\zeta_X(t)$ represents the normalized seasonal transmission pattern in island X , and $\alpha_X = (\max(\beta_X(t)) - \min(\beta_X(t)))/2$ represents the amplitude of seasonal transmission pattern in island X . This allowed us to interpolate between two normalized seasonal terms and obtain a flexible shape for seasonal transmission rate:

$$\zeta_{\text{new}}(t) = \zeta_H(t)(1 - \theta) + \zeta_K(t)(\theta), \quad (10)$$

$$\beta_{\text{new}} = \left(\frac{\alpha_{\text{new}} \zeta_{\text{new}}(t)}{(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2} + 1 \right) \bar{\beta}_{\text{new}}, \quad (11)$$

where θ represents the interpolation coefficient, such that $\theta = 0$ and $\theta = 1$ causes β_{new} to have the same shape as β_H and β_K , respectively and $0 < \theta < 1$ allows us to model counterfactual transmission scenarios that interpolates between two islands. Note that $\zeta_{\text{new}}(t)$ does not necessarily have an amplitude 1 so we divide it by $(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2$ to ensure the amplitude of 1.

For a given value of the interpolation coefficient θ and seasonal amplitude α_{new} , we simulated two outbreaks for a year assuming $s(0) = 0.078$ and $s(0) = 0.105$ and computed the difference in the timing of epidemic peak. In doing so, all other parameters, including the mean transmission rate β_{new} , were fixed to posterior median estimates for the Honshu island.

Regression

We performed univariate quadratic regressions for the estimated transmission rates against mean specific humidity and mean temperature, separately, and computed R squared values for each regression. We also perform bivariate quadratic regressions for the estimated transmission rates using both mean specific humidity and mean temperature as covariates. The significance of regression coefficients was determined using two-sided t-tests at the 0.05 significance level.

We performed a linear regression between the estimated proportion of susceptibles at the beginning of each season (week 26) against the number of children attending childcare facilities. For simplicity, the regression was performed using median estimates for the susceptible proportions. We did not have data on the number of children attending childcare facilities broken down by island level and so we used the national-level data instead.

404 Data availability

405 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parksw3/perturbation)
406 [parksw3/perturbation](https://github.com/parksw3/perturbation)) [?].

407 Code availability

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493 Acknowledgements

494 We acknowledge the efforts of the National Institute of Infectious Diseases, Statis-
495 tics Bureau of Japan, and Children and Families Agency for collecting/maintaining
496 the data used in this study and making them publicly available. E.H., B.G., and
497 C.J.E.M. have been funded in whole or in part with Federal funds from the Na-
498 tional Cancer Institute, National Institutes of Health, under Prime Contract No.
499 75N91019D00024, Task Order No. 75N91023F00016. The content of this publica-
500 tion does not necessarily reflect the views or policies of the National Institutes of
501 Health or the Department of Health and Human Services, nor does mention of trade
502 names, commercial products or organizations imply endorsement by the U.S. Govern-
503 ment. S.W.P acknowledges support from Peter and Carmen Lucia Buck Foundation
504 Awardee of the Life Sciences Research Foundation and the New Faculty Startup Fund
505 from Seoul National University. I.H. received postdoctoral funding from the High
506 Meadows Environmental Institute of Princeton University. B.T.G. and C.J.E.M.
507 acknowledge support from Princeton Catalysis Initiative and Princeton Precision
508 Health. S.C. is supported by Federal funds from the National Institute of Allergy
509 and Infectious Diseases, National Institutes of Health, Department of Health and
510 Human Services under CEIRR contract 75N93021C00015 - Subcontract 77789. The
511 content is solely the responsibility of the authors and does not necessarily represent
512 the official views of the NIAID or the National Institutes of Health.

513 Contributions

514 S.W.P., I.H., and B.T.G. conceived of the study. S.W.P. performed the analysis and
515 wrote the initial draft. All authors (S.W.P., I.H., E.H., W.Y., R.E.B., G.A.V., S.C.,
516 C.J.E.M., and B.T.G.) reviewed and edited the manuscript.

517 Competing Interests

518 The authors declare no competing interests.

519 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ($n=52$). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ($n=52$). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ($n=8000$ posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ($n=8000$ posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $S(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$. (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from $n=8000$ posterior samples. Shaded regions represent the 95% credible interval in our estimates ($n=8000$ posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from $n=8000$ posterior samples. Error bars represent the 95% credible interval in our estimates ($n=8000$ posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.